

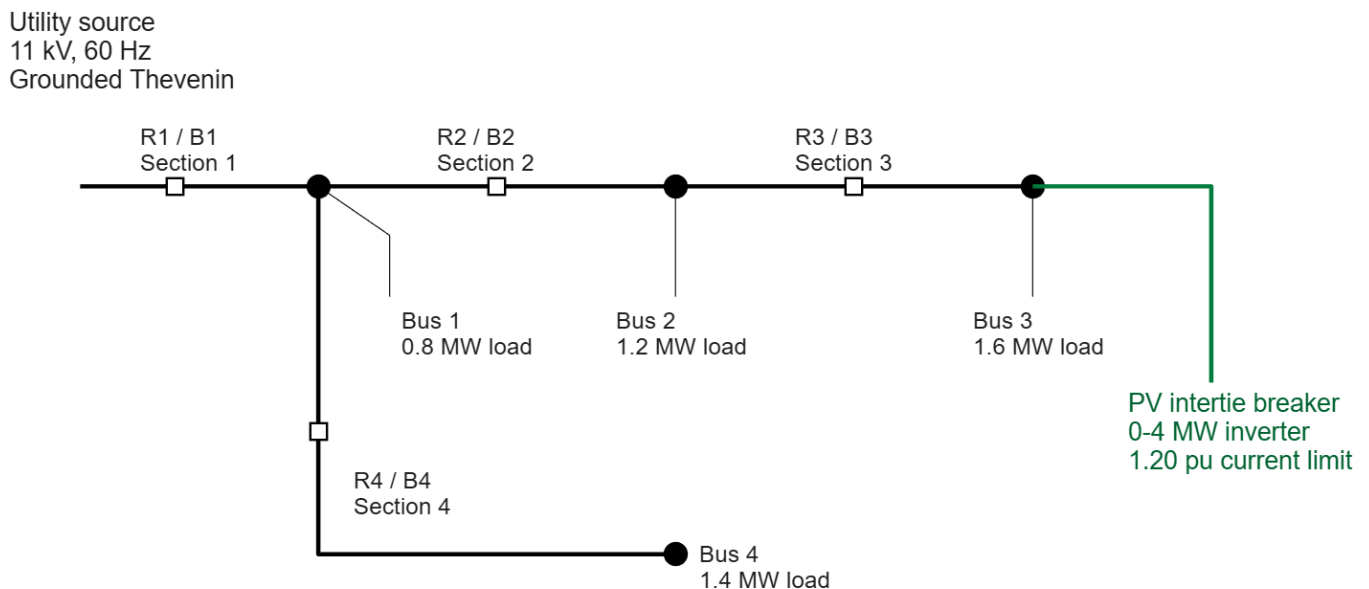
Protection coordination with distributed solar — engineering report

Generated by the executed MATLAB study. MATLAB 25.2.0.2998904 (R2025b); seed 20261004.

1. Objective and network

Design and test selective feeder protection with current-limited solar generation. The 11 kV, 60 Hz radial feeder has four protected sections, 5 MW + 1.25 Mvar of nominal fixed-impedance load, and 0–4 MW of PV at bus 3. Solar penetration is installed PV divided by the 5 MW nominal active load: 0–80%.

Distribution feeder single-line diagram



Faults at 10%, 50%, and 90% of each section; faulted line is split at the fault node.
R1 backs up R2 and R4; R2 backs up R3. Opening B1/B2/B3 initiates direct PV transfer trip.

Single-line diagram

Source positive/negative-sequence impedance is $0.12 + j1.20$ ohm; zero sequence is $0.25 + j1.80$ ohm. Source impedance scales from 0.8 to 1.5 in the design envelope. Line zero-sequence impedance is three times positive sequence. The source is grounded, loads are grounded wye, and inverter injection has positive sequence only. See `src/study_config.m` for every network parameter.

2. Electrical and inverter models

This is a three-phase RMS/phaser protection study, not an electromagnetic-transient or switching-inverter model. A complex 15-node phase-domain nodal admittance solve computes voltages, CT-location line currents, and fault currents. Each faulted line is split at its specified fault location. Breaker opening removes

its sending-end connection; the downstream section and inverter remain electrically represented.

The inverter uses an ideal source-angle reference and voltage-dependent active/reactive current commands. Reactive current receives priority during voltage sag; total phase current is limited to 1.20 times installed rated current. Available active output is installed rating times irradiance. The solved current command depends on actual terminal voltage. There is no detailed DC link, PLL, negative-sequence control, switching ripple, or vendor-specific fault response. This ideal synchronized current-source abstraction can continue energizing an island until its intertie opens; direct transfer trip is explicitly modeled rather than assuming instant solar disconnection.

3. Protection settings and coordination

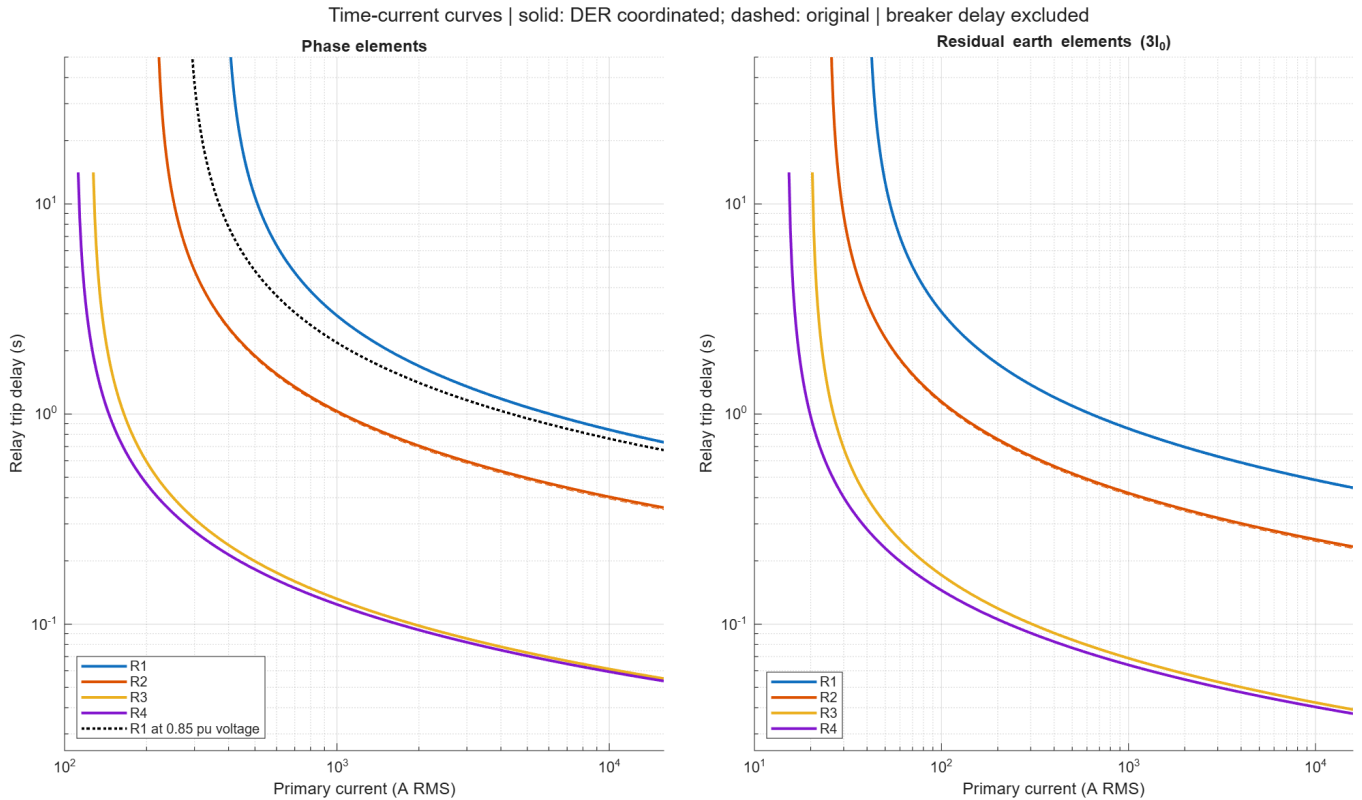
The relay has independent operate accumulators for phases A/B/C and residual current 3I0. The IEC normal-inverse equation is $t = TMS * 0.14 / ((I/I_{pickup})^{0.02} - 1)$ above pickup. Each enabled channel integrates dt/t ; dropout resets its accumulator and a trip latches. Current is specified in primary RMS amps; CT ratios and equivalent 1 A-secondary settings are tabulated below.

Conventional 51/51N uses no direction supervision. The directional scheme adds phase-by-phase memory-voltage polarization with a 60 degree characteristic angle and zero-sequence voltage polarization for ground direction. Phase memory uses the ideal nominal source reference; this abstraction is not a full commercial relay polarization algorithm. The improved scheme additionally restrains R1 phase pickup using $\max(0.5, \min(1, V_{min_pu}^2))$ and coordinates TMS across the full DER envelope. R1 voltage restraint addresses weak source-current backup sensitivity found at high solar penetration.

Forward load-envelope phase pickup has 20% reserve and is rounded upward in 5 A steps. Residual pickup has 50% reserve with explicit minimum settings. Each relay shares one TMS across its four channels. Fixed pickups and a radial relay hierarchy make minimum-TMS grading transparent: solve R2 after R3, then R1 after R2/R4. Each bound is $TMS_{up} \geq (K_{down} * TMS_{down} + 0.25 + 0.05 + 0.001) / K_{up}$, where K is the minimum enabled unit-TMS channel time. TMS is rounded upward in 0.005 steps. This is constrained grading of fixed pickups, not a global optimization of all possible relay designs.

The 0.25 s required coordination margin is measured from primary breaker opening to the prospective backup relay command, with 0.05 s breaker time and a 1 ms allowance. R1 backs up R2/R4; R2 backs up R3; R1 provides another level of backup for R3. Opening B1/B2/B3 sends a direct PV transfer trip with 20 ms communication delay and 40 ms PV-intertie breaker time. B4 opening leaves PV grid-connected. No automatic reclosing is modeled.

Relay	CT ratio	Phase pickup A primary	Residual pickup A primary	Baseline TMS	Improved TMS	51V
R1	600/1	385	40	0.400	0.405	1
R2	300/1	215	25	0.225	0.230	0
R3	300/1	125	20	0.040	0.040	0
R4	200/1	110	15	0.040	0.040	0



4. Test design

The design envelope contains 7200 cases: four faulted sections, ten fault types, locations 0.1/0.5/0.9 of each section, fault resistances 0.1/2/10 ohm, PV ratings 0/1/2/3/4 MW, two source-impedance scales, and two load scales (0.25 and 1.2). Load unbalance is 3%. ABC denotes a balanced three-phase-to-ground fault; ABG/BCG/CAG are explicit interphase-plus-ground conductance networks.

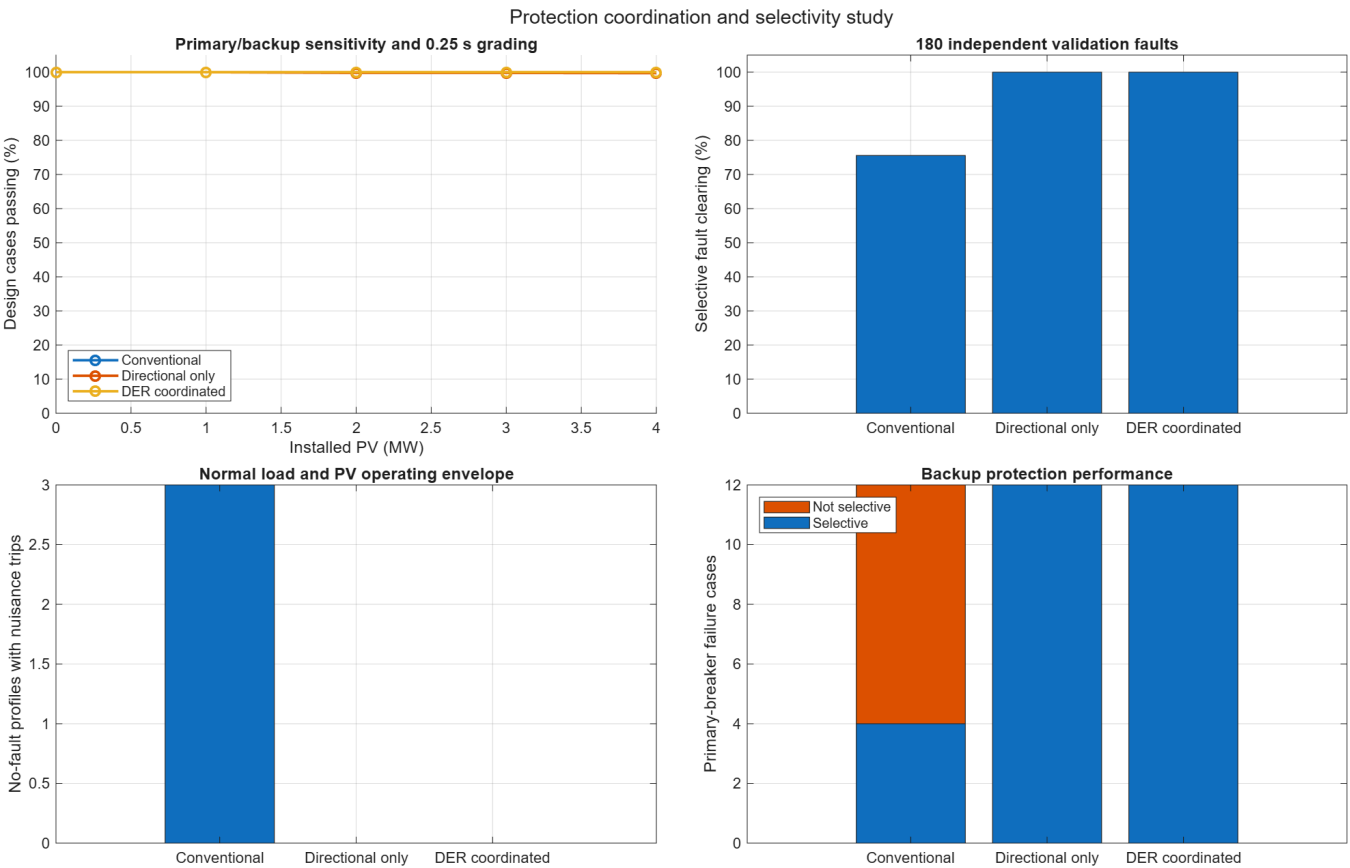
Independent validation uses 180 newly randomized scenarios with continuously varying fault location, resistance, PV rating, irradiance, source impedance, load scale, and unbalance. These scenarios are not used to set TMS. Dynamic evaluation includes 360 randomly selected design cases, the independent validation cases, no-fault operating profiles, primary-breaker failure cases, and three intentionally disabled-transfer-trip cases. Every scenario runs all three protection schemes.

Event-driven simulation exactly integrates stationary RMS channel rates between circuit-changing events. The live Simulink model separately runs at 1 ms and feeds breaker states back into the circuit with a unit delay. Neither simulation includes a waveform-derived phasor measurement delay. Results are simulation results, not field or hardware validation.

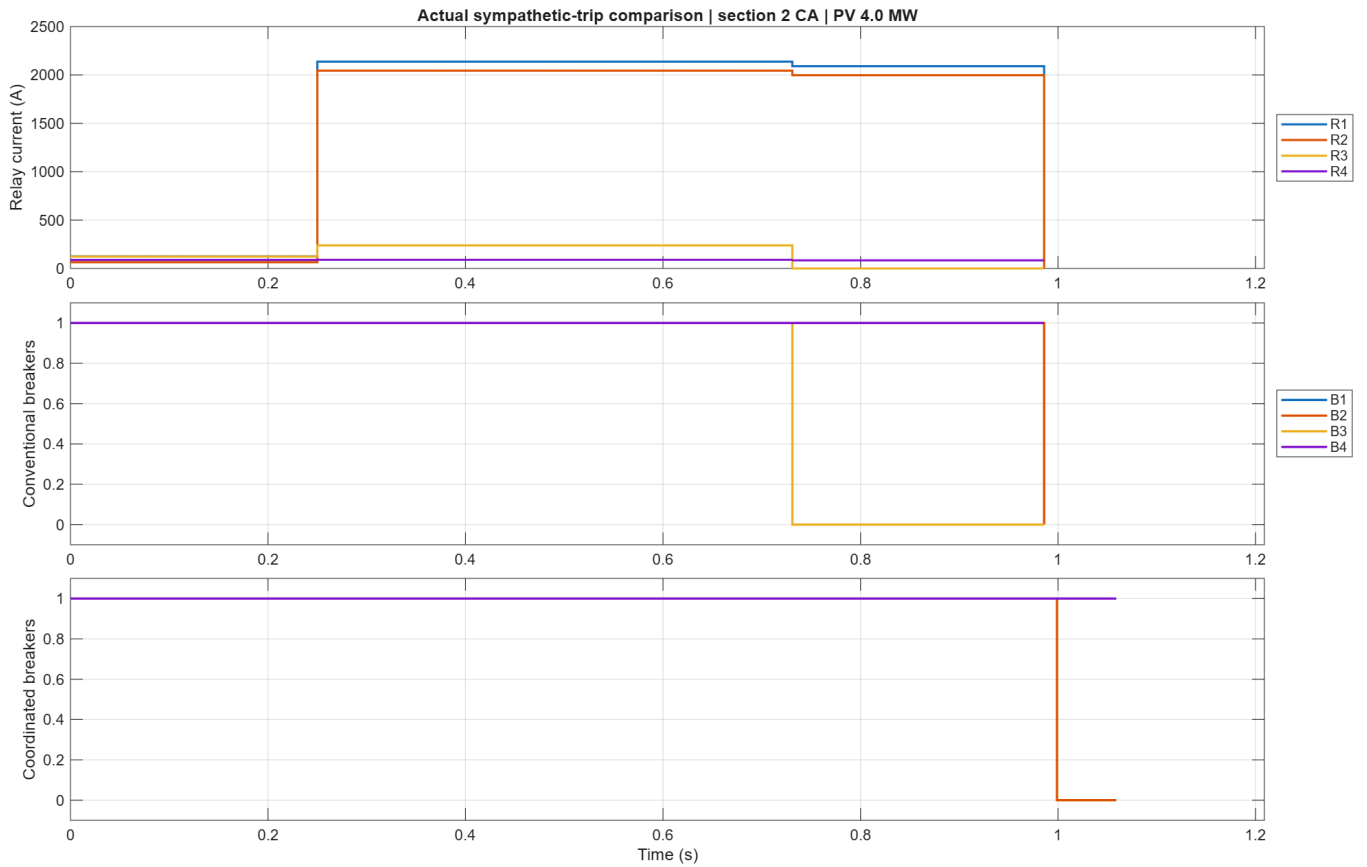
5. Executed results

Scheme	Design grading violations	Missing required relay	Minimum finite margin s	Independent selective faults	No-fault nuisance cases	Selective breaker-failure cases
Conventional 51/51N	10/7200	1	0.2455	136/180	3/15	4/12
Directional 67/67N	10/7200	1	0.2455	180/180	0/15	12/12
DER-coordinated 67/67N + 51V	0/7200	0	0.2534	180/180	0/15	12/12

A missing required primary/backup relay is assigned negative-infinite grading margin; it is a failed sensitivity case, not a successful infinitely long delay. Cases with only R1 as primary have no in-model upstream backup and are excluded from minimum finite pair margin.



Validation outcomes



Sympathetic-trip comparison

The default live Simulink demonstration is a midpoint section-3 AG fault at $R_f=2$ ohm, with 4 MW PV. The event-driven reference clears it 0.1769 s after inception. B3 opens first; PV can continue contributing until its inertia opens following transfer trip. See [results/demo_event_log.csv](#) and [results/simulink_traces.csv](#) for actual timings and traces.



Live Simulink results

6. Verification and practical limits

Independent Thevenin and sequence-network formulas verify balanced and AG fault currents with no loads/PV. All design cases meet nodal KCL residual and inverter current-limit checks. Improved design cases meet 0.25 s grading. Dynamic tests verify no-fault behavior, fault isolation, primary-breaker backup, sustained inverter contribution when transfer trip is disabled, and a two-breaker-failure source-backup case. Every live Simulink measurement is checked against a separate circuit solve; relay/breaker timings agree with the event-driven reference within two 1 ms steps. Exact checks are in `results/verification.txt`.

The maximum primary trip delay in the improved design envelope is 10.728 s, compared with 17.151 s for the conventional settings. High-resistance, near-pickup faults can therefore remain slow despite meeting grading and sensitivity requirements. This project imposes no universal maximum-clearing-time requirement. A practical study would need equipment damage curves, acceptable fault-duration limits, more sensitive elements or communication-assisted protection where required, and manufacturer-specific settings validation.

Further limits: ideal CTs/VTs, no CT saturation, no harmonics, no arc model, fixed-impedance loads, no transformer energization, no finite relay memory decay, no neutral conductor model, and a simplified inverter. The fault-resistance range is not a validation of high-impedance arcing-fault detection. Meeting this study envelope does not certify compliance with an IEC or IEEE standard.

The severe two-breaker-failure sensitivity check requires about 31 s for source backup. This is an intentional diagnostic of near-pickup inverse-time delay, not a claim of acceptable utility protection speed. Communication-assisted or differential protection would need investigation where equipment fault-duration limits rule out such delayed backup. Six additional saved-model scenarios verify breaker failure, conventional sympathetic trips, directional selectivity, no-fault reverse export, source-zone isolation and disabled transfer trip.

7. Sources

- MathWorks: overcurrent relay coordination in an AC microgrid
- NREL: protection challenges with inverter-based resources
- NREL report: Understanding Fault Characteristics of Inverter-Based Distributed Energy Resources
- SEL: voltage-supervised and voltage-restrained overcurrent elements — background only; the project uses a custom restraint characteristic.